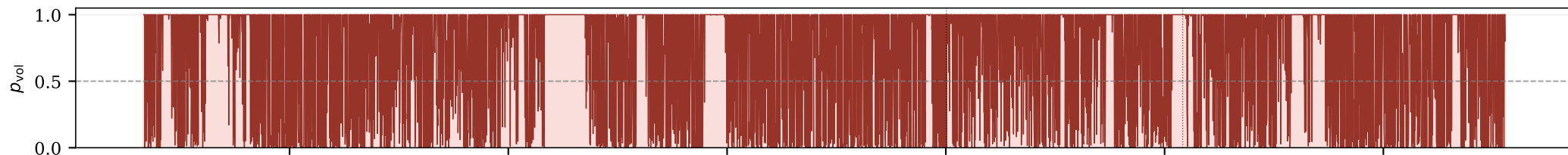


HMM-posterior p_{volatile} across the full timeline, by HMM variant. O vs Os: same features, different HMM emission model.

Variant O HMM (GMMHMM mix=2 BIC winner, 11 features) **TRAIN**

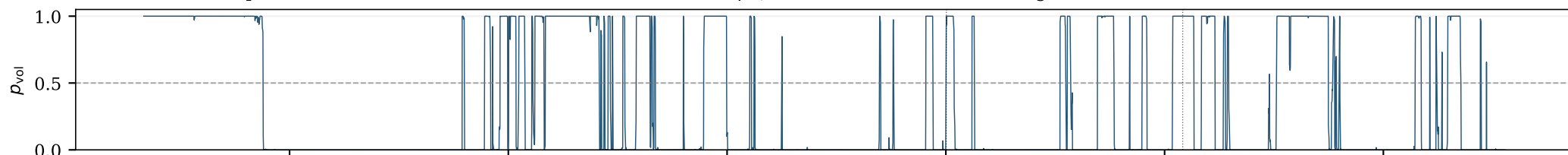
test: mean $p_{\text{vol}}=0.67$, share $> 0.5=67\%$, lag-1 autocorr=0.19

VAL **TEST**



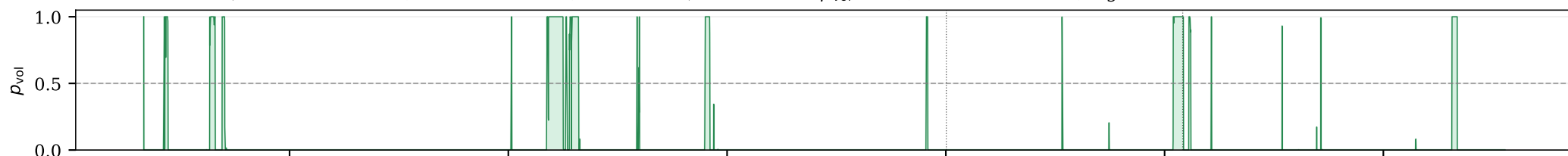
Variant Os HMM (plain GaussianHMM, same 11 features)

test: mean $p_{\text{vol}}=0.33$, share $> 0.5=33\%$, lag-1 autocorr=0.97



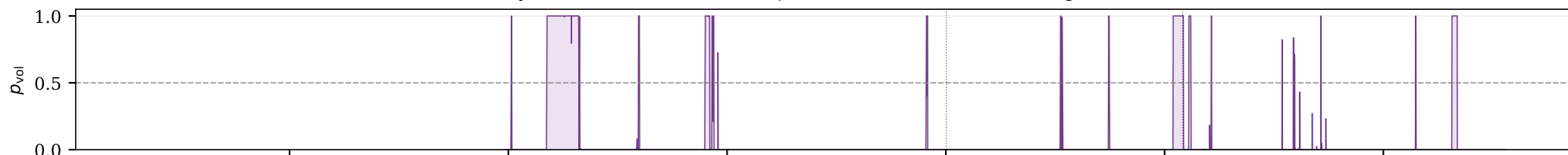
Variant A HMM (GMMHMM mix=2, +AAII sentiment, 13 features)

test: mean $p_{\text{vol}}=0.03$, share $> 0.5=3\%$, lag-1 autocorr=0.87



Variant B HMM (GMMHMM mix=2, +VIX-family, 16 features)

test: mean $p_{\text{vol}}=0.03$, share $> 0.5=3\%$, lag-1 autocorr=0.82



2004

2008

2012

2016

2020

2024

Date (full timeline: train + val + test)